

WEEKLY PROGRESS REPORT

Rice Leaf Disease Recognition and Classification
Reproduction Study: EfficientNet-B0 CNN From Scratch

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Programme	B.Tech – Artificial Intelligence
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Project Stage	Week 1 Progress
Reporting Date	05 October 2026
Scope of this report	Research-paper study → Dataset → CNN architecture from scratch → CNN baseline

1. Executive Summary

This week's work focused on studying the selected research paper and reproducing its CNN component as a standalone baseline. The dataset was organized and verified, preprocessing and DataLoaders were tested, the MBConv building block was implemented and verified, and an EfficientNet-B0 style CNN was constructed from scratch. The network was then trained for 8 epochs using AdamW and Cross-Entropy Loss. The best observed test accuracy of the scratch CNN was **99.47%** at Epoch 7.

A key methodological point is that the **original research paper uses pretrained models**: it uses a pretrained EfficientNet-B0 CNN and a pretrained ViT-Small model (ImageNet pretraining). In this student project, the architecture is being rebuilt and trained **from scratch**. Therefore, our results are an independent scratch-training experiment and should be compared with the paper's reported results as a reference, not presented as an exact reproduction of the paper's pretrained experiment.

2. Short Summary of the Actual Research Paper

Paper: "Efficient-ViT B0Net: A high-performance lightweight transformer for rice leaf disease recognition and classification" by Santosh Kumar Upadhyay and Rajesh Prasad, Journal of the Nigerian Society of Physical Sciences, Volume 7 (2025), Article 2940.

The paper addresses automated rice-leaf disease recognition using a hybrid CNN–Vision Transformer architecture. The motivation is that CNNs are strong at extracting local and fine-grained texture patterns, whereas Vision Transformers can model broader/global relationships between image features. The proposed Efficient-ViT B0Net therefore combines EfficientNet-B0 for compact local feature extraction with ViT-Small for transformer-based global representation.

The study uses a rice-leaf dataset containing four classes: Bacterial Leaf Blight, Brown Spot, Health, and Leaf Smut, with 4,000 images per class. The dataset is divided into 3,200 training and 800 testing images per class (80:20 split). Images are processed at 224×224 resolution. In the hybrid architecture, EfficientNet-B0 produces a 7×7×320 feature map, which is flattened into 49 tokens and projected to 384 dimensions before entering the ViT component.

The paper reports a proposed-model testing accuracy of **99.13%**, with macro Precision, Recall and F1-score also reported as 99.13%. Its comparison table reports 91.3% for EfficientNet-B0 and 97.0% for ViT-Small. The paper also discusses the advantage of reducing the number of transformer tokens through CNN downsampling, thereby reducing the computational burden of self-attention.

3. Critical Methodological Difference: Paper vs. Our Project

Component	Original Research Paper	Our Project
EfficientNet-B0	Pretrained CNN architecture	Implemented from scratch
ViT-Small	Pretrained on ImageNet	Implemented from scratch
Main purpose of CNN	Local/fine-grained feature extraction	Same architectural role, learned from scratch
CNN output	7×7×320 feature map	7×7×320 feature map
Hybrid connection	49×320 → 49×384 → ViT	Planned for next stage
Training	Paper's reported experimental setup	Our own scratch-training experiment

Why this distinction matters: A pretrained model starts with features learned from a large external dataset, while our scratch model starts with randomly initialized trainable weights and must learn the image features from the rice-leaf dataset itself. Consequently, matching the paper's exact accuracy is not expected or required for this stage.

4. Project Objective

The project objective is to study the selected paper, understand its architecture, and progressively implement the model. The planned final workflow is: **Dataset → EfficientNet-B0 from scratch → standalone ViT-Small from scratch → EfficientNet-B0 + ViT-Small hybrid → comparison with the paper.**

This weekly report deliberately stops after the standalone CNN stage. The ViT and hybrid stages will be documented in subsequent progress reports.

5. Dataset and Data Preparation

Class	Total	Training	Testing
Bacterial Leaf Blight	4,000	3,200	800
Brown Spot	4,000	3,200	800
Health	4,000	3,200	800
Leaf Smut	4,000	3,200	800
Total	16,000	12,800	3,200

Input images were resized to **224×224** and converted into PyTorch tensors. The DataLoader was verified using batch size 32. A training batch had shape **[32, 3, 224, 224]** and labels had shape **[32]**.

6. Work Completed This Week

Dataset verification

Confirmed the four classes and 80:20 class-balanced train/test organization.

Preprocessing

Verified 224×224 image resizing and tensor conversion.

DataLoader

Verified batch size, image tensor shape, label shape, and input range.

MBConv block

Implemented and independently tested the Mobile Inverted Bottleneck Convolution block.

EfficientNet-B0

Constructed the CNN architecture from scratch without loading pretrained EfficientNet weights.

Feature-map verification

Verified the required $7 \times 7 \times 320$ intermediate CNN representation.

Training setup

Configured GPU execution, Cross-Entropy Loss and AdamW with learning rate 0.001.

CNN baseline

Completed 8-epoch training and recorded epoch-wise loss and accuracy.

7. EfficientNet-B0 Architecture Implemented From Scratch

Stage	Channels	Spatial size	Role
Input	3	224×224	RGB image
Stem	32	112×112	Initial convolution
MBConv Stage 1	16	112×112	Early local features
MBConv Stage 2	24	56×56	Downsampling
MBConv Stage 3	40	28×28	Texture/pattern extraction
MBConv Stage 4	80	14×14	Higher-level features
MBConv Stage 5	112	14×14	Feature refinement
MBConv Stage 6	192	7×7	High-level features
MBConv Stage 7	320	7×7	Hybrid feature-map target

The key representation required by the paper's hybrid design is $7 \times 7 \times 320$. In the next stage, this will be reshaped to 49×320 tokens and projected to 49×384 before entering the ViT-Small encoder.

8. CNN Training Configuration

Parameter	Value
Model	EfficientNet-B0 from scratch
Input	224×224×3
Batch size	32
Epochs	8
Optimizer	AdamW
Learning rate	0.001
Loss	Cross-Entropy Loss
Execution	Google Colab GPU (NVIDIA T4)

9. EfficientNet-B0 Scratch Training Results

Epoch	Train Loss	Train Acc.	Test Loss	Test Acc.
1	0.4246	82.88%	0.2191	92.16%
2	0.1579	94.55%	0.5759	84.72%
3	0.1019	96.65%	0.0517	98.69%
4	0.0811	97.17%	0.8647	85.41%
5	0.0678	97.81%	0.2648	91.38%
6	0.0407	98.62%	0.4379	85.09%
7	0.0938	97.30%	0.0114	99.47%
8	0.0318	99.02%	0.0255	99.03%

Best scratch CNN result: 99.47% test accuracy at Epoch 7. The final Epoch 8 test accuracy was 99.03%.

10. Comparison With the Main Research Paper Results

The following comparison is included to clearly distinguish the paper's reported benchmarks from the results obtained in our scratch implementation.

Model / Result	Research paper	Our project (this week)	Difference / interpretation
EfficientNet-B0	91.3%	99.47% best test accuracy	Our scratch result is higher, but the setups are not identical; this is not a fair comparison.
ViT-Small	97.0%	Not included this week	ViT stage will be reported separately after scratch implementation and comparison.
Proposed hybrid EfficientNet-ViT	99.80%	Not yet implemented	Hybrid is the next project stage.

Important interpretation: The paper's EfficientNet-B0 and ViT-Small benchmarks are reported in the paper's experimental environment, while our current CNN is a from-scratch implementation. Differences in initialization, pretraining, implementation details, augmentation, optimization and checkpoint selection can affect accuracy. The comparison is therefore used to understand the gap between the reference work and our implementation, not to claim an exact reproduction.

11. Discussion

The CNN baseline demonstrates that the implemented MBConv-based architecture can learn the four-class rice disease problem effectively. The best test accuracy of 99.47% was obtained at Epoch 7. The variation in test accuracy across epochs shows why epoch-wise evaluation and recording are important. The retained $7 \times 7 \times 320$ feature map is also technically important because it will become the compact token source for the final hybrid model.

The main research contribution that will be reproduced in the next stage is the combination of local CNN features and global transformer attention. The paper's design uses CNN downsampling to reduce the transformer token count, with $7 \times 7 \times 320$ converted to 49×320 and then to 49×384 . This is the connection that will be implemented after the standalone ViT experiment.

12. Skills and Concepts Learned

- Research-paper architecture reading and workflow decomposition
- Dataset organization and class-wise verification
- PyTorch ImageFolder/DataLoader pipeline
- Image resizing and tensor preprocessing
- MBConv implementation and testing
- EfficientNet-B0 architecture construction without pretrained weights

- GPU execution in Google Colab
- Cross-Entropy Loss and AdamW optimization
- Forward propagation and backpropagation
- Epoch-wise evaluation and baseline comparison

13. Next Week's Planned Work

- Implement the standalone ViT-Small completely from scratch.
- Train the ViT-Small baseline and record accuracy/loss curves.
- Compare EfficientNet-B0 and ViT-Small standalone results.
- Build the $7 \times 7 \times 320 \rightarrow 49 \times 320 \rightarrow 49 \times 384$ hybrid interface.
- Combine the scratch EfficientNet-B0 and scratch ViT-Small end-to-end.
- Train and evaluate the final hybrid architecture.
- Compare the final model with the research paper's 99.13% hybrid result.

14. References

1. Upadhyay, S. K., & Prasad, R. (2025). *Efficient-ViT B0Net: A high-performance lightweight transformer for rice leaf disease recognition and classification*. Journal of the Nigerian Society of Physical Sciences, 7, 2940. DOI: 10.46481/jnsps.2025.2940.
2. The selected paper's reported dataset, architecture, training configuration, comparison results and proposed hybrid workflow were used as the reference basis for this weekly implementation.
3. Project implementation: PyTorch/Google Colab implementation developed during the current project work. The EfficientNet-B0 baseline results reported in this document were obtained from the student's scratch implementation.

Appendix A — Key Tensor Shapes

Operation	Tensor shape
Input image batch	[32, 3, 224, 224]
EfficientNet-B0 feature map	[32, 320, 7, 7]
CNN classifier output	[32, 4]